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Krylov-space anatomy and spread complexity of a disordered quantum spin chain

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We investigate the anatomy and complexity of quantum states in Krylov space, in the ergodic and many-body localised (MBL) phases of a disordered, interacting spin chain. The Krylov basis generated by the Hamiltonian from an initial state provides a representation in which the spread of the time-evolving state constitutes a basis-optimised measure of complexity. We show that the long-time Krylov spread complexity sharply distinguishes the two phases. In the ergodic phase, the infinite-time complexity scales linearly with the Fock-space dimension, indicating that the state spreads over a finite fraction of the Krylov chain. By contrast, it grows sublinearly in the MBL phase, implying that the long-time state occupies only a vanishing fraction of the chain. Further, the profile of the infinite-time state along the Krylov chain exhibits a stretched-exponential decay in the MBL phase. This behaviour reflects a broad distribution of decay lengthscales, associated with different eigenstates contributing to the long-time state. Consistently, a large-deviation analysis of the statistics of eigenstate spread complexities shows that while the ergodic phase receives contributions from almost all eigenstates, the complexity in the MBL phase is dominated by a vanishing fraction of eigenstates, which have anomalously large complexity relative to the typical ones.

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I. INTRODUCTION

The many-body localisation transition [1, 2] in disordered quantum many-body systems can be viewed as an eigenstate transition: across a critical disorder strength, the nature of eigenstates at arbitrary finite energy densities changes qualitatively between the ergodic and many-body localised (MBL) phases (see Refs. [3–9] for reviews and further references). One of the most prominent real-space signatures of this transition is encoded in the eigenstate expectation values of local observables. In the ergodic phase, these satisfy the eigenstate thermalisation hypothesis [10–14], whereas in the MBL phase they violate it.

From a purely eigenstate perspective, the distinction between the ergodic and MBL phases is also reflected in bipartite entanglement entropy. Eigenstates in the ergodic phase typically exhibit volume-law entanglement, while those in the MBL phase obey an area law [3, 5, 15–17]. The latter is intimately connected to the existence of an extensive number of (quasi)local integrals of motion [16–18]. Consequently, MBL eigenstates can be related to trivial product states by finite-depth local unitary circuits, whereas no such local circuit exists for ergodic eigenstates [15]. This perspective suggests that eigenstates in the two phases possess fundamentally different complexities, at least in the abstract sense of circuit complexity [19–21]

A complementary and equally insightful approach to the many-body localisation problem exploits the exact mapping of any disordered interacting many-body Hamiltonian to a tight-binding Hamiltonian, describing a fictitious single particle hopping on the associated correlated Fock-space graph of the system; where sites correspond to many-body basis states and links correspond to matrix elements of the Hamiltonian [22–30] (see Refs. [7, 8] for reviews of this approach). The structure of many-

body eigenstates on this graph is qualitatively different in the ergodic and MBL phases, as reflected e.g. in their inverse participation ratios (IPRs). Specifically, MBL eigenstates display multifractal statistics, manifest in anomalous scaling of the IPRs with the Fock-space dimension $N_{\mathcal{H}}$, while ergodic eigenstates are fully extended over the graph [26, 27, 31–34]. Physically, this implies that MBL eigenstates occupy a number of Fock-space sites which, although exponentially large in system size, constitutes a vanishing fraction of the total Fock-space dimension, while ergodic eigenstates by contrast occupy a finite fraction of the graph.

A dynamical manifestation of this distinction arises when the system is initialised in a many-body basis state corresponding to a single site of the Fock-space graph. Under time evolution, the state spreads over the graph. In the ergodic phase, the long-time state occupies a finite fraction of the graph, whereas in the MBL phase it remains confined to a vanishing fraction even at arbitrarily late times [35, 36]. This suggests the existence of a notion of complexity associated with the spread of the state on the Fock-space graph, which should behave qualitatively differently in the two phases.

An intuitive measure of this complexity is the size of the wavefunction on the graph. But such a measure is basis dependent, so this naturally raises the question of whether there exists a basis in which the spread of the state – and hence the associated complexity – is minimised. Importantly, it was shown that the Krylov basis generated by the Hamiltonian from the initial state minimises the complexity of the time-evolving state [37]. Consequently, the spread complexity of the state in Krylov space (hereafter referred to as the Krylov spread complexity) provides a bona fide, basis-optimised measure of complexity.

This in turn motivates the central question of this work: what is the anatomy of states in Krylov space in the ergodic and MBL phases, and how does the Krylov spread complexity distinguish between them? We also show that the Krylov-space anatomy of states provides insights into the MBL phase that are arguably more difficult to obtain from the Fock-space perspective. This is because the Hamiltonian in Krylov space takes the form of a one-dimensional (correlated) disordered tight-binding chain of length $N_{\mathcal{H}}$, where the ordering of basis states and the notion of distance are much simpler than on the high-dimensional Fock-space graphs.

The present work should be contrasted with the much more widely studied operator Krylov complexity [38–51], where the operator growth of an initially real-space-local operator is mapped onto the spread of the operator in the Krylov space of the operator Hilbert space. In fact, the operator Krylov complexity is not necessarily a sensitive probe of the MBL phase, because in this phase too the operator may delocalise in the Krylov space, as shown e.g. for the phenomenological ℓ-bit model [43] (neither in fact is it a sensitive probe of integrability [52]).

On the other hand, as shown in this work, the Krylov-space anatomy, as well as the Krylov spread complexity of an initial state [53–63] evolved for arbitrarily long (infinite) times, shows qualitative differences in the ergodic and MBL phases. In fact, the Krylov spread complexity has already proven successful in capturing aspects of ergodicity breaking in both single-particle and many-body settings under specific initial conditions [53–59]. More specifically, we find that the infinite-time Krylov spread complexity in the ergodic phase scales linearly with the Fock-space dimension, $\propto N_{\mathcal{H}}$, whereas in the MBL phase it grows sublinearly as $N_{\mathcal{H}}^{\alpha}$ with $\alpha < 1$. Coupled to the fact that the Krylov space is an ordered one-dimensional chain of length $N_{\mathcal{H}}$, the above scaling presents a simple yet insightful picture: in the ergodic phase the infinite-time states extend over a finite fraction of the entire Krylov chain, while in the MBL phase, they extend to only a vanishing fraction of the chain.

In the MBL phase we find in fact that the profile of the infinite-time state along the length of the Krylov chain has a non-trivial decay described by a stretched exponential. We present a phenomenological theory which shows how such a profile can emerge, from the interplay of exponential decay of the state amplitude on the Krylov chain and a broad (but not itself heavy-tailed) distribution of associated decay length scales. Importantly, these distributions arise not only over disorder realisations, but also across eigenstates within a given realisation. This interplay becomes evident when the infinite-time Krylov spread complexity is expressed as a sum over eigenstates of the Krylov spread complexities associated with individual eigenstates.

The latter decomposition yields further insight into the nature of eigenstates in the two phases. In the ergodic phase, eigenstate spread complexities are found to have a narrow distribution over the eigenstates. By contrast, in the MBL phase the distribution is broad, and its tails become heavier with increasing system size. Using a large-deviation analysis, we show that in the ergodic phase the infinite-time Krylov spread complexity receives contributions from almost all eigenstates, whereas in the MBL phase it is dominated by only a vanishing fraction of eigenstates – albeit exponentially many in system size – lying in the tails of the distribution. This provides a manifestation, on the Krylov chain, of rare resonances characteristic of the MBL phase [28, 64–74].

The paper is organised as follows. Sec. II reviews briefly the definition of the Krylov spread complexity and the construction of the Krylov basis. In Sec. III, we specify the disordered, tilted-field Ising chain as our model of choice, and discuss its Krylov chain and its conceptual connections to the model's Fock-space graph. Detailed numerical results for the Krylov-space anatomy and the spread complexity, along with a phenomenological theory in the MBL phase, constitute Sec. IV. The large-deviation analysis of the statistics of the eigenstate spread complexities contributing to the infinite-time spread complexity is presented in Sec. V. Some concluding remarks are given in Sec. VI.

II. KRYLOV SPREAD COMPLEXITY

We start by discussing the construction of the Krylov chain, and formally defining the associated spread complexity [37]. This enables us to discuss several important conceptual points, and to put on a concrete footing the motivation behind studying the anatomy of states on the Krylov chain.

Consider a general many-body Hamiltonian of form

$$H = H_0 + H_1 \,, \tag{1}$$

denoting by $\{|I\rangle\}$ the eigenstates of H_0, and by $\{|E\rangle\}$ the eigenstates of H with eigenvalue E. The setting in which we will be interested is initialising the state of system at $t = 0$ in an eigenstate of H_0, $|\psi_{t=0}\rangle = |I\rangle$, and asking how *complex* the state becomes upon evolution over an arbitrarily long time under the full Hamiltonian $|\psi_t\rangle = e^{-iHt}|\psi_0\rangle$. Denoting by $S_K(t)$ the measure of complexity for the state at time t, our specific focus will be the long-time limit $S_{K,\infty} \equiv \lim_{t\to\infty} S_K(t)$. This can be expressed in terms of correlated eigenstate amplitudes on the Krylov chain, as we show below.

For a general state $|\psi_t\rangle$, given some orthonormal basis $\{|\mathcal{V}_n\rangle\}$, this can be quantified using a cost function

$$C_{\mathcal{V}} = \sum_n \mu_n |\langle\psi_t|\mathcal{V}_n\rangle|^2 \,, \tag{2}$$

where the μ_ns are a positive, increasing sequence of real numbers. It is customary [37] to consider $\mu_n = n$, such that $C_{\mathcal{V}}$ can be interpreted as the size of the support of the state $|\psi_t\rangle$ assuming the basis $\{|\mathcal{V}_n\rangle\}$ to be ordered. A natural measure of complexity is therefore to consider the basis which minimises the cost function. An important result in this context is that the basis which minimises Eq. 2 is the Krylov basis [37], generated via the Hamiltonian H, with its first basis vector given by $|\psi_0\rangle$. This is an ordered orthonormal basis, the basis states being denoted by $|k_n\rangle$ with $n = 0, 1, \cdots, N_{\mathcal{H}} - 1$, where $N_{\mathcal{H}}$ is the Fock-space dimension of the system. These Krylov basis states are defined via the recursion relation [75]

$$|k_n\rangle = \frac{1}{b_n}\left[H|k_{n-1}\rangle - a_{n-1}|k_{n-1}\rangle - b^*_{n-1}|k_{n-2}\rangle\right] , \tag{3}$$

where $a_n = \langle k_n|H|k_n\rangle$ and $b_n = \langle k_n|H|k_{n-1}\rangle$, and $|k_0\rangle = |I\rangle$ is the boundary condition for the recursion.

An immediate consequence of this construction is that the many-body Hamiltonian in the Krylov basis takes the tridiagonal form

$$H = \sum_{n=0}^{N_{\mathcal{H}}-1} a_n |k_n\rangle\langle k_n| + \sum_{n=1}^{N_{\mathcal{H}}-1} \left[b_n |k_{n-1}\rangle\langle k_n| + \text{h.c.}\right] . \tag{4}$$

The Krylov basis is thus equivalent to a one-dimensional, nearest-neighbour, tight-binding chain of length $N_{\mathcal{H}}$, described by the Hamiltonian Eq. 4. While the fact that the length of the Krylov chain is $N_{\mathcal{H}}$ is a straightforward fallout, its conceptual importance is that it provides us with a naturally ordered one-dimensional basis for the many-body Hilbert space, which has a simple and unambiguous notion of distance.

With the initial state $|I\rangle = |k_0\rangle$ localised at one end of the chain, the temporal evolution of the state on the Krylov chain is encoded in the probability amplitudes

$$c_n(t) = \langle k_n|\psi_t\rangle = \sum_E e^{-iEt}\langle k_n|E\rangle\langle E|k_0\rangle \,. \tag{5}$$

The spread complexity is then defined as [37]

$$S_K(t) = \sum_{n=0}^{N_{\mathcal{H}}-1} n|c_n(t)|^2 \,, \tag{6}$$

and the one-dimensional nature of the Krylov chain leads to the interpretation of the spread complexity as simply the size of the support of the wavefunction on the chain.

We will be particularly interested in the spread complexity at infinite time, formally defined as

$$S_{K,\infty} = \lim_{t\to\infty} \frac{1}{t}\int_0^t dt' \; S_K(t') \,. \tag{7}$$

This can be conveniently expressed as

$$S_{K,\infty} = \sum_{n=0}^{N_{\mathcal{H}}-1} n\Lambda_n \,, \tag{8}$$

where (with $|E\rangle$ denoting an eigenstate of H)

$$\Lambda_n = \sum_E \Lambda_{n,|E\rangle} \text{ with } \Lambda_{n,|E\rangle} = |\langle k_n|E\rangle\langle E|k_0\rangle|^2 \,. \tag{9}$$

Equations 8 and 9 relate the infinite-time spread complexity to eigenstate amplitudes on the Krylov chain. Note that conservation of probability trivially leads to Λ_n satisfying the sum rule

$$\sum_{n=0}^{N_{\mathcal{H}}-1} \Lambda_n = 1 \,. \tag{10}$$

As such, Λ_n can interpreted as a probability distribution over the Krylov chain, for which $S_{K,\infty}$ is nothing but its first moment. The profile of Λ_n on the Krylov chain, and its scaling with $N_{\mathcal{H}}$, together with the statistical properties of the eigenstate spread complexity $S_{K,\infty}$, will be the central quantities of interest in this work.

III. DISORDERED SPIN-1/2 CHAIN AND ITS KRYLOV CHAIN

A. Model

As a concrete setting for our studies, we consider the disordered tilted-field Ising (TFI) spin-1/2 chain, which

has emerged as the standard archetype for a disordered system hosting a MBL phase at strong disorder [8, 9, 76, 77]. The model is described by the Hamiltonian

$$H = \underbrace{\sum_{i=1}^{L-1} J_i\, \sigma_i^z \sigma_{i+1}^z + \sum_{i=1}^{L} h_i \sigma_i^z}_{H_0} + \underbrace{\Gamma \sum_{i=1}^{L} \sigma_i^x}_{H_1}, \tag{11}$$

where (taking $\Gamma \equiv 1$) $J_i \in [0.8, 1.2]$ and $h_i \in [-W, W]$ are independent random numbers. This model hosts an MBL phase for strong enough disorder, and an ergodic phase at weaker disorder strengths. For the parameters above, within accessible system sizes, the critical disorder strength is $W \simeq 3.7$ [77].

Given that H_0 in Eq. 11 is composed solely of $\{\sigma_i^z\}$ operators, the basis $\{|I\rangle\} = \{|\{\sigma_i^z\}\rangle\}$ is simply the σ^z-product state basis. The 0th Krylov orbital, $|k_0\rangle$, is then simply a specific σ^z-product state, which we denote by $|I_0\rangle$. In our numerical calculations we consider $|I_0\rangle$ to be the σ^z-product state whose energy is closest to the middle of the many-body spectrum. With this choice, the entire set of Krylov-basis vectors $\{|k_n\rangle\}$ together with the parameters $\{a_n\}$ and $\{b_n\}$ describing the Krylov Hamiltonian, can be computed explicitly. Some details of the statistical properties of the $\{a_n, b_n\}$ are relegated to Appendix A, but one essential point should be mentioned here. Specifying an instance of the Hamiltonian in Eq. 11 requires only $\propto L$ independent random values of J_i and h_i. By contrast, specifying the Krylov Hamiltonian requires $\propto N_{\mathcal{H}}$ values of a_n and b_n (Eq. 4). However, since the latter $\propto N_{\mathcal{H}}$ coefficients are constructed from only $\propto L$ independent J_i and h_i values, strong correlations between the matrix elements of the Krylov Hamiltonian necessarily arise. As a result, the problem is very different from a conventional Anderson localisation problem [78–80], despite the Krylov chain being one dimensional.

B. On the connection between Krylov and Fock-space bases

Before delving into the anatomy of eigenstates on the Krylov chain, it is important from a conceptual point of view to understand the connection between the Krylov basis and the more commonly employed Fock-space basis, and its associated Fock-space graph.

The eigenstates of H_0 (σ^z-product states $\{|I\rangle\}$) form a natural choice of basis for the Fock space. Properties of the Fock-space graph have been discussed in detail previously (see [8] and references therein); here we summarise the features of it essential for the present work.

The Fock-space graph for the TFI chain, illustrated as part of Fig. 1, is an L-dimensional hypercube with $N_{\mathcal{H}} = 2^L$ nodes, each such representing a σ^z configuration. Each node has a connectivity of precisely L, which corresponds to flipping each of the L spins on the chain

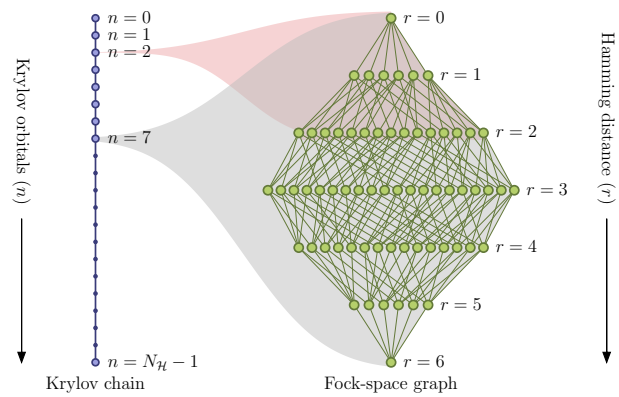

FIG. 1. In the Krylov-space basis, the many-body Hamiltonian takes the form of a nearest-neighbour tight-binding chain (Eq. 4), of length $N_{\mathcal{H}} = 2^L$, with the orbitals $|k_n\rangle$ labelled $n = 0, 1, \cdots, N_{\mathcal{H}} - 1$. On the Fock-space graph (illustrated for $L = 6$), the Krylov $|k_0\rangle$ orbital is identified with the top apical site $|I_0\rangle$. For $n \leq L$, $|k_n\rangle$ is supported on Fock-space sites which lie within Hamming distance $r \leq n$, as illustrated by the red-shaded connection between $|k_2\rangle$ and all Fock-space sites with $r \leq 2$. By contrast, for $L + 1 \leq n \leq N_{\mathcal{H}} - 1$, $|k_n\rangle$ is in general supported on the entire Fock-space graph, as indicated by the grey-shaded connections.

under the action of H_1 ($= \Gamma \sum_{i=1}^{L} \sigma_i^x$). The Hamming distance between two σ^z configurations – defined as the number of sites where the two configurations differ from each other – endows the Fock-space graph with a natural notion of distance. For the form of H_1 in Eq. 11 it is also, conveniently, the shortest path between the two nodes. The above features imply that for a given $|I_0\rangle$, the remaining nodes can be arranged in a row-wise fashion such that all nodes on row r are a Hamming distance of r from $|I_0\rangle$. Naturally there are L such rows, with the rth row containing $N_r = \binom{L}{r}$ nodes. The form of H_1 in Eq. 11 also implies that any node on row r is connected to nodes only on rows $r \pm 1$.

Given the structure of the Fock-space graph, a natural question is: what is the distribution of the Krylov-basis vectors, $\{|k_n\rangle\}$, on the Fock-space graph? To make this quantitative, define the overlap of the Krylov-basis states and the Fock-space basis states as

$$w_{nI} = |\langle k_n | I \rangle|^2, \tag{12}$$

and the average weight of the nth Krylov-basis state at Hamming distance r from I_0 on the Fock-space graph as

$$w_n(r) = \frac{1}{N_r} \sum_{I: r_{II_0} = r} w_{nI} = \frac{1}{\binom{L}{r}} \sum_{I: r_{II_0} = r} |\langle k_n | I \rangle|^2. \tag{13}$$

Considering $|k_0\rangle = |I_0\rangle$, the construction of the Krylov chain in Eq. 3, together with the form of H_1, makes it clear that for $n \leq L$, the Krylov $|k_n\rangle$ has support on Fock-space sites which lie at Hamming distances $r \leq n$ from I_0. However, for $n > L$, the orthonormality of $|k_n\rangle$ with

$|k_m\rangle$ $\forall\, m < n$ implies that the support of $|k_n\rangle$ folds back on the Fock-space graph, and thus in principle exists on the entire graph. This is shown schematically in Fig. 1 for the case of $L = 6$. Indeed, since the Krylov chain has a length $N_{\mathcal{H}} = 2^L$, a fraction tending to unity of the sites thereon potentially have support over the entire Fock space.

It is of course important to establish that the Krylov-basis states indeed span the Fock-space graph, whether in the ergodic or MBL regimes. To this end, we define an effective size of the wave function corresponding to the Krylov orbital $|k_n\rangle$ on the Fock-space graph, via

$$\mathcal{R}_n = \sum_{r=0}^{L} r \binom{L}{r} w_n(r) = \sum_{r=0}^{L} r \sum_{I: r_{I I_0} = r} w_{nI} \,. \tag{14}$$

Physically, $\mathcal{R}_n$ gives the mean Hamming distance on the Fock-space graph that is associated with the given Krylov $|k_n\rangle$, and as such is a bona fide measure of how the Krylov-basis state is spread over the Fock-space graph. Since $\{|k_n\rangle\}$ forms a complete orthonormal basis, such that $\sum_n w_n(r) = 1$ (Eq. 13), note that $\mathcal{R}_n$ satisfies a sum rule

$$\sum_{n=0}^{N_{\mathcal{H}}-1} \mathcal{R}_n = \sum_{r=0}^{L} r \binom{L}{r} = N_{\mathcal{H}} L/2 \,. \tag{15}$$

Sufficiently deep in the ergodic phase, one expects $w_{nI} \sim N_{\mathcal{H}}^{-1}$ and hence $w_n(r) \sim N_{\mathcal{H}}^{-1}$ for all I and $n > L$; which when used in Eq. 14 yields $\mathcal{R}_n \simeq L/2$. That this is indeed the case is confirmed by numerical results shown in Fig. 2(a) for a disorder strength $W = 1$ deep in the ergodic phase (and with system sizes L as indicated).

The natural generalisation of this to encompass the MBL regime is the binomial form

$$w_n(r) = p_n^r (1 - p_n)^{L-r} \,, \tag{16}$$

which from Eq. 14 yields $\mathcal{R}_n = p_n L$ (with the deep ergodic phase corresponding to $p_n = 1/2$). The sum rule in Eq. 15 then implies $\sum_n p_n = N_{\mathcal{H}}/2$. Assuming $p_n = g(n/N_{\mathcal{H}}^a)$ to be some function of $n/N_{\mathcal{H}}^a$ (with exponent a), this sum rule can be cast for $N_{\mathcal{H}} \gg 1$ as

$$\int_0^{N_{\mathcal{H}}} dn\; g\left(\frac{n}{N_{\mathcal{H}}^a}\right) = N_{\mathcal{H}}^a \int_0^{N_{\mathcal{H}}^{1-a}} dx\; g(x) = \frac{N_{\mathcal{H}}}{2} \,, \tag{17}$$

which naturally requires $a = 1$. This then implies that $\mathcal{R}_n$ satisfies the scaling form

$$\mathcal{R}_n = L \times g\left(\frac{n}{N_{\mathcal{H}}}\right) \,, \tag{18}$$

such that $\mathcal{R}_n/L$ is a function solely of $n/N_{\mathcal{H}}$. While the scaling form is quite trivially satisfied in the ergodic regime (as above), numerical results in the MBL phase, shown in Fig. 2(b) for disorder strength $W = 10$, confirm this to be case in general. The key upshot of this scaling form is that in both the ergodic and MBL regimes, the Krylov-basis states indeed have access to the entire Fock-space graph [1].

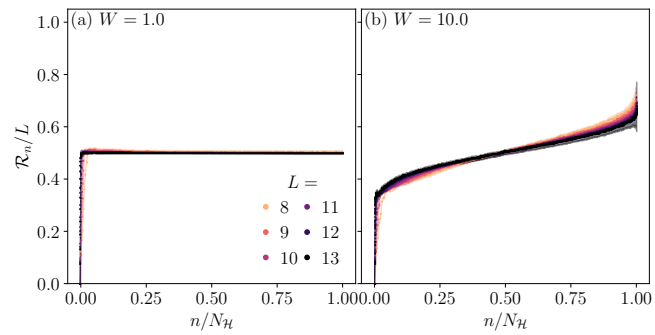

FIG. 2. Numerical results for the effective size of the wave function on the Fock-space graph corresponding to the nth Krylov orbital, as quantified by $\mathcal{R}_n$ defined in Eq. 14 (with system sizes L indicated). Panels (a) and (b) with $W = 1$ and $W = 10$ exemplify, respectively, the ergodic and MBL regimes. The results conform to the scaling form in Eq. 18. The (thin) shaded region around the data points reflects the statistical errors over disorder realisations.

IV. SCALING FORMS FOR SPREAD COMPLEXITY AND EIGENSTATE AMPLITUDES

With the basic properties of the Krylov chain and the spread complexity defined, we turn now to the Krylov-space anatomy of the eigenstates.

A. Distribution and scaling of $S_{K,\infty}$

We start with the distribution of the infinite-time spread complexity over disorder realisations; in particular the distribution $P_s(s)$ of the late-time complexity rescaled by its mean, $s = S_{K,\infty}/\langle S_{K,\infty}\rangle$, numerical results for which are shown in Fig. 3 for representative W's in each of the ergodic and MBL phases.

In the ergodic regime the distributions are well described by a Gaussian with mean 1

$$P_s(s) = \frac{1}{\sqrt{2\pi\sigma_s^2}} \exp\left[-\frac{(s-1)^2}{2\sigma_s^2}\right] \,, \tag{19}$$

and clearly narrow rapidly with increasing system size. Results for the mean $\langle S_{K,\infty}\rangle$ are given in Fig. 4(a), from which it is evident that in the ergodic regime,

$$\langle S_{K,\infty}\rangle \simeq N_{\mathcal{H}}/2 \,. \tag{20}$$

[1] This is not of course the case for the first L Krylov basis states, but that is immaterial since these constitute an exponentially small (in L) fraction of the 2^L Krylov states.